Enhanced Image Inpainting and Visual Prompting Using Masked Autoencoders and Adversarial Training

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Abstract
Masked Autoencoder (MAE) is a pre-trained model designed to learn visual representations by reconstructing masked image patches. While MAE excels in image recognition tasks, its potential for generative tasks remains underexplored. To fill this gap, we propose integrating MAE into a Generative Adversarial Network (GAN), and incorporating perceptual loss to enhance the clarity and visual quality of inpainted images. We extended this enhanced architecture, termed MAE-GAN, beyond image inpainting to broader visual prompting tasks. By employing a one-shot-like approach, MAE-GAN generalizes its utility to virtually any visual task. Our work demonstrates the effectiveness of MAE's visual representations and extends its application to generative tasks.

I. Introduction

Representation learning methods have recently gained popularity in computer vision, particularly for their ability to pre-train models that are effective in downstream image tasks. Visual representation learning, in this context, focuses on learning generalizable features from large-scale datasets. MAE [1] presents a simple yet effective method for visual representation learning: it masks random patches from the input image and reconstructs missing patches in the pixel space. This approach has demonstrated strong generalization capabilities for downstream tasks: a vanilla Vision Transformer [2] (ViT-Huge) model pre-trained using MAE achieves an 87.8% accuracy on ImageNet-1K [1] dataset. However, MAE research has primarily focused on image recognition tasks, such as classification, segmentation, and objection detection, leaving its potential in image generation largely unexplored.

Despite its strengths, MAE exhibits certain limitations when applied to image generation: (1) the Mean Squared Error (MSE) loss often leads to blurry output images that lack sharp details and textures; (2) due to MAE's patch-based reconstruction approach, visual inconsistencies often appear at the boundaries of adjacent patches, disrupting the coherence of the image; and (3) discrepancies can arise between the generated results and the unmasked regions of the original image, further affecting the reconstruction quality. These artifacts collectively prevent the visual appeal of images generated by MAE and limit its applicability in high-fidelity image generation tasks.

Most artifacts in MAE-generated images can be attributed to perceptual inconsistencies. We extend MAE by integrating it as the generator within a GAN-like architecture, which we term MAE-GAN. The key idea behind Generative Adversarial Networks (GANs) [3] is the simultaneous training of two networks, a generator, in our case MAE, and a discriminator in a competitive manner. The discriminator enhances the generator by rejecting sub-optimal reconstructions, such as blurry images. Through this competitive training process, the discriminator learns to assess whether an image is visually plausible, while the generator (MAE) learns to produce more visually convincing outputs.

In addition to the GAN architecture, we apply perceptual loss [4] to further enhance the perceptual quality of the generated images. It measures the difference in high-level features extracted from pre-trained convolutional neural networks (CNNs), such as VGG16 [5], when real and generated images are passed as inputs. By computing the MSE loss between the feature maps produced at various layers of the pre-trained network, perceptual loss captures content, texture, and style discrepancies between images, which helps to generate images that are more perceptually appealing.

Apart from improving MAE's image inpainting capabilities, we expand its potential by fine-tuning MAE-GAN on visual prompt-guided image inpainting tasks [6]. Inspired by one-shot learning in Natural Language Processing (NLP), this technique emphasizes the model's ability to learn patterns and perform inference with limited examples. During test time, the model is provided with an input-output image example (or examples) of a new task, along with a new input image. The goal is for the model to generate an output image consistent with the given example(s). In our case, an input-output image example, a query input, and a blank image are combined into a single image (as shown in Fig. 2) and passed through MAE-GAN. The blank region in the combined image is reconstructed by MAE-GAN to complete the task.

Our main contributions can be summarized as follows:

1) We propose the integration of MAE into a GAN-like architecture, introducing a novel model called MAE-GAN.
2) We recognized the primary limitation of MAE as being unable to generate clear, detailed images, and designed tailored loss functions to tackle this issue.
3) We trained MAE-GAN on image inpainting tasks and demonstrated through qualitative and quantitative evaluations, that it generates images with higher

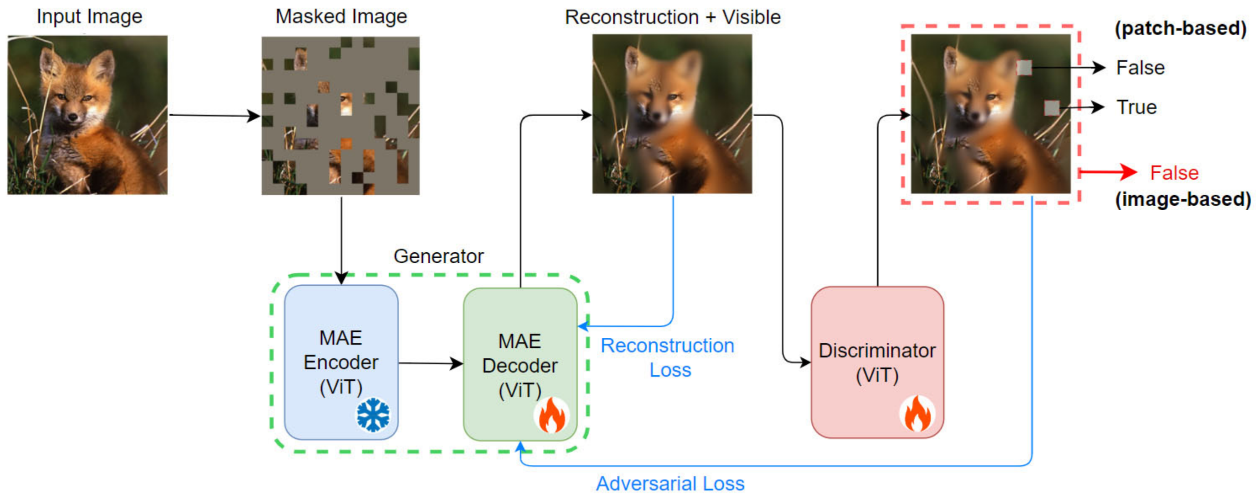

Fig. 1. **MAE-GAN architecture**. An MAE serves as the generator, tasked with reconstructing images. A discriminator distinguishes not only between real and generated images but also between real and generated image patches.

clarity and fewer artifacts compared to MAE.

4) We extended the capabilities of MAE-GAN using visual prompting techniques, showcasing its multi-task potential through qualitative assessments.

II. Related Work

Masked Autoencoders (MAE). MAE [1] is a self-supervised model designed to learn robust image representations by reconstructing randomly masked patches of an input image. Although MAE is primarily intended as a general-purpose pre-trained model, its patch reconstruction capability makes it particularly suitable for tasks like image inpainting.

GAN. GAN [3] refers to a class of machine learning model architecture that consists of two competing neural networks: generator and discriminator in the form of a zero-sum game. The generator learns to produce data aligned with the distribution of the training set, while the discriminator learns to distinguish between real and generated data. During training, both models are dynamically updated, enabling the generator to iteratively improve the quality of its outputs. GAN and its variants have been widely applied in image generation. Conditional GANs [7] enable conditional image generation by incorporating conditional embeddings into the architecture. Similarly, Oord et al. [8] introduced PixelCNN [9] into GANs to facilitate text-guided image generation.

GAN-MAE. Recent work [10] has explored the combination of MAE and GAN architectures. The MAE-GAN model integrates MAE's reconstruction capabilities with the adversarial learning dynamics of GANs to enhance the sharpness and quality of generated images. Our project builds on this by incorporating perceptual loss to preserve high-frequency information and introducing boundary loss to address the boundary artifacts that emerge from the MAE's patch-based reconstruction. Specifically, the boundary loss calculates the difference between pixels in adjacent image patches to enforce smoother transitions and reduce visual inconsistencies.

Perceptual Loss. Image content and style are not always evident at the pixel level. The idea behind perceptual loss [4] is to leverage feature maps extracted from a pre-trained CNN (e.g. VGG16) to measure differences between generated and target images in terms of high-level features, such as edges, textures, and patterns. By comparing feature maps across multiple layers, perceptual loss ensures that the generated images align with the visual semantics of the target.

Visual Prompting. Visual prompting [6] enables models to generalize tasks based on input-output examples provided at test time. This paradigm is particularly effective for image restoration, where the model learns to infer patterns from pairs of corrupted and complete images provided during testing. We seek to apply this paradigm to MAE-GAN. By combining MAE's patch reconstruction capabilities with adversarial refinement, we adapt visual prompting to enable the model to generalize across tasks such as colorization and deblurring.

III. Method

MAE-GAN employs an MAE as the generator and a vision transformer as the discriminator, forming a GAN structure for image inpainting. The model incorporates enhanced loss functions to minimize unwanted artifacts. In this section, we first explain the various components of MAE-GAN in detail and then describe our approaches to extend MAE-GAN's utility beyond image inpainting by visual prompting.

A. MAE

The MAE architecture comprises two components: an encoder with a Vision Transformer (ViT) backbone and a ViT-based lightweight decoder. Given an input image

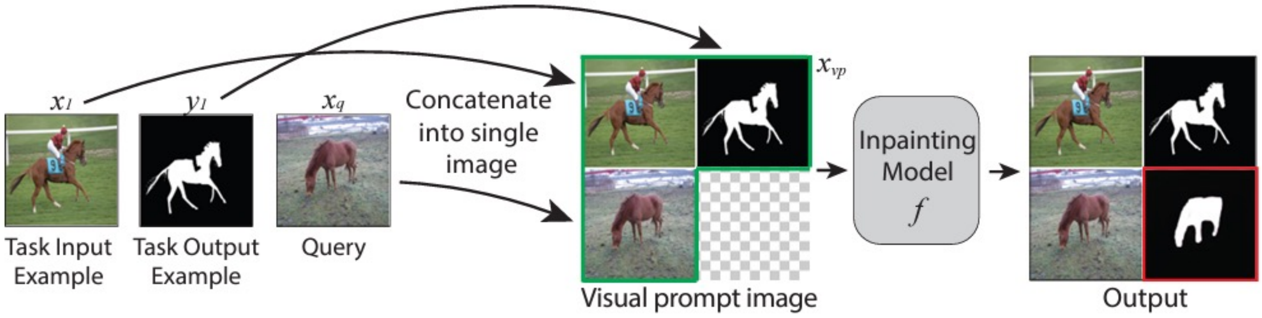

Fig. 2. **Image prompting.** The input image is divided into four quadrants, and each quadrant is a sub-image. [6] The upper-left quadrant is the task input example, the upper-right quadrant is the task output example, the bottom-left quadrant is the query image, and the bottom-right quadrant is the blank image to be reconstructed by an image model.

$$X \in \mathbb{R}^{C \times H \times W}$$

where C, H, and W are the channel number, image height, and image width, respectively. The image is then divided into $N = \frac{H \times W}{P^2}$ non-overlapping rectangular patches of size $P \times P$. This results in a sequence of patches $\{x_1, \cdots, x_N\}$, where each patch $x_i \in \mathbb{R}^{C \times P^2}$. MAE then applies a random masking strategy by selecting a subset of patch indices M, masking a large proportion (typically 75%) of patches. The remaining visible patches are fed into the encoder, which produces compact high-level image representations. To reconstruct the full image, learnable mask tokens are inserted at the positions of corresponding masked patches respectively, and the resulting sequence is passed to the decoder. The decoder predicts each image patch $\{\hat{x}_1, \cdots, \hat{x}_N\}$ in the pixel space. For the image-inpainting task, only the reconstruction of the masked patches $\{\hat{x}_i\}_{i \in M}$ is relevant. MAE minimizes the mean squared error reconstruction loss between the predicted patches and the ground truth patches over the masked regions:

$$L_{recon} = \sum_{i \in M} ||\hat{x}_i - x_i||_2^2$$

Since the reconstruction process does not account for smooth transitions at the edges of neighboring patches, and the MAE is not explicitly trained to generate photorealistic images with high fidelity, the resulting images often appear blurry and exhibit inconsistencies both between adjacent patches and between reconstructed patches and the image context.

B. MAE-GAN Discriminator

To suppress blurriness and patch inconsistency in the predicted images, we integrate MAE into the GAN architecture where MAE acts as the generator. The ViT-based discriminator is trained to learn whether a prediction produced by the MAE is visually plausible or not as shown in Fig. 1. The discriminator, denoted as D, aims to facilitate the MAE generator by rejecting suboptimal reconstruction.

The MAE-predicted masked patches $\{\hat{x}_i\}_{i \in M}$ are pasted back onto the unmasked patches $\{x_i\}_{i \notin M}$ to obtain a merged image $\hat{X}$. We believe that providing the discriminator D with the merged image is beneficial compared to providing only the predicted patches since D can learn the valuable context from the unmasked portion of the ground truth that is otherwise discarded.

With a single ViT backbone, the discriminator plays two roles. Firstly, the class token within ViT targets the entire image and judges whether the entire image is real or fake. The loss function for discriminating the entire image is:

$$L_{\text{disc\_image}} = -\log D(X) - \log(1 - D(\hat{X}))$$

Secondly, each patch token within the ViT outputs whether the corresponding image patch is real or fake. The loss function for discriminating each image patch is:

$$L_{\text{disc\_patch}} = \sum_{i=1}^{N} -\log D(x_i) - \log(1 - D(\hat{x}_i))$$

The MAE-GAN discriminator minimizes the combined loss function:

$$L_{\text{disc}} = \alpha_1 \cdot L_{\text{disc\_image}} + \alpha_2 \cdot L_{\text{disc\_patch}}$$

where α_1, α_2 are hyperparameters.

C. MAE-GAN Generator

Similar to the discriminator losses, the adversarial losses for the generator are:

$$L_{\text{adv\_image}} = -\log(1 - D(\hat{X})),$$
$$L_{\text{adv\_patch}} = \sum_{i=1}^{N} -\log(1 - D(\hat{x}_i))$$

GAN is prone to unstable training, which refers to the circumstance where the generator or discriminator dominates the other, resulting in poor convergence and limiting the model's learning ability. To address this challenge in MAE-GAN, we incorporated a perceptual loss to stabilize

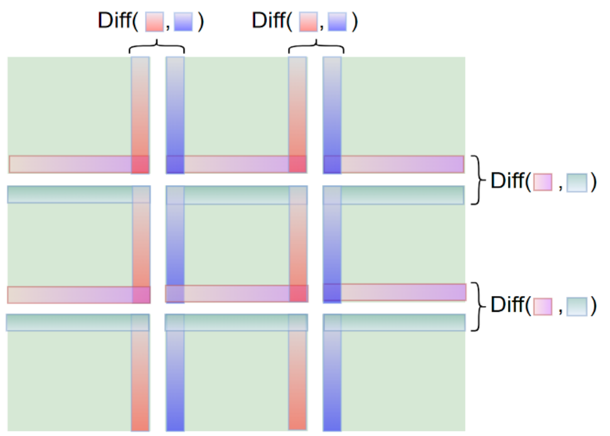

Fig. 3. **Visualization of Boundary Loss**. Minimizing the differences among patches' edges is treated as the optimization objective.

adversarial training and improve the quality of generated images.

The ground truth X and generated images $\hat{X}$ are fed into a pre-trained image model F. Intermediate features from different layers of this model are then extracted and compared. The perceptual loss is:

$$L_{\text{percep}} = ||F(\hat{X}) - F(X)||_2^2$$

By focusing on high-level feature similarity rather than pixel-wise differences, the perceptual loss ensures that the generator produces outputs that align closer with human perception. This reduces over-reliance on the discriminator's feedback, which can sometimes lead to undesired artifacts.

To tackle the patch inconsistency problem, we need to reduce inconsistency between neighboring patches in MAE-generated images. Explicitly, we penalize pixel-wise differences along patch boundaries. For each patch boundary, the pixel values on one side are subtracted from those on the other. This leads to the boundary loss as shown in Fig. 3:

$$L_{\text{bound}} = \sum_{i=1}^{H} \sum_{n=1}^{\frac{W}{P}-1} ||\hat{X}_{i,(n\cdot P)} - \hat{X}_{i,(n\cdot P+1)}|| + \sum_{m=1}^{\frac{H}{P}-1} \sum_{j=1}^{W} ||\hat{X}_{(m\cdot P),j} - \hat{X}_{(m\cdot P+1),j}||$$

where $\hat{X}_{i,j}$ represents the pixel value on row i, column j of the generated image. The boundary loss proved effective in enforcing patch-patch consistency as well as image-mask consistency.

The overall loss function for the MAE generator is:

$$L_{\text{gen}} = L_{\text{recon}} + \alpha_1 \cdot L_{\text{adv\_image}} + \alpha_2 \cdot L_{\text{adv\_patch}} + \alpha_3 \cdot L_{\text{percep}} + \alpha_4 \cdot L_{\text{bound}}$$

where $\alpha_1, \cdots, \alpha_4$ are hyperparameters.

D. Image prompting

Visual prompting extends the utility of MAE-GAN beyond image inpainting to a broader range of visual tasks. Given an input-output image example of a new task at test time and a new input image, the goal of visual prompting is to automatically produce the output image, consistent with the given example.

The task is specified using an input-output example (x_l, y_l), where x_l is the input image, and y_l is the corresponding output image. Along with this example, a query image x_q and a blank image are used to construct a single image. These components are arranged into four quadrants as shown in Fig. 2. The blank part of the combined image represents the masked region to be reconstructed. This combined image is fed into MAE-GAN, which interprets the relationship between x_l and y_l to determine the transformation required. It then reconstructs the blank region to generate the output corresponding to x_q.

IV. Experiments

A. Implementation Details

ViT Backbones. Throughout the experiments, we used a pre-trained ViT-Large backbone for the MAE-GAN generator and a pre-trained ViT-Small backbone for the MAE-GAN discriminator. Both ViT models use patch size 16. The pre-trained weights for the MAE were obtained from [1], while the pre-trained weights for the discriminator were obtained from the PyTorch timm library ("vit_small_patch16_224"). For computational efficiency, the MAE encoder is frozen, and we fine-tuned only the MAE decoder and the discriminator.

Dataset. The models were fine-tuned using $13,676$ images from the PASCAL 2012 [11] dataset. For MAE-GAN fine-tuning, all images were resized to 224×224, and 75% of the image patches were randomly masked to simulate missing regions. To fine-tune MAE-GAN for visual prompting, we constructed a custom dataset. Each image y_l is paired with another randomly sampled image y_q from the same dataset. We randomly applied one of the following transformations to both images in the pair: Grayscaling, Random Square Masking, and Gaussian Blurring. For Random Square Masking, the mask ratio is 10%. For Gaussian blurring, the kernel size and sigma were set to 5×5 and 2.0, respectively. Both images (y_l, y_q) and their transformed versions (x_l, x_q) were resized to 112×112 and combined into a single image of size 224×224 as shown in Fig. 2. The bottom-right quadrant of the combined image corresponds to the desired output y_q, which is masked, resulting in a mask ratio of 25%.

Parameter Configuration. The batch size was set to 32 for all experiments. We used Adam [12] optimizer and cosine scheduling for model training. We fine-tuned each MAE-GAN for 10 epochs and the visual prompting model for 15 epochs. Each model took 60 to 80 minutes to train on an NVIDIA RTX-2060 GPU.

TABLE I
HUMAN PREFERENCE RESULTS FOR THREE INPAINTING MODELS. "MAE-GAN WITH LOSS" REFERS TO THE MODEL TRAINED WITH OUR CUSTOM-DESIGNED LOSS FUNCTION. THE VALUES ARE REPORTED AS PERCENTAGES, AND HIGHER IS BETTER. THE BEST ENTRIES ARE BOLDED.

Model Architecture	Overall	Grid Artifact	Blurriness Artifact	Image Patch Artifact
MAE	7.22	16.67	0.56	5.56
MAE-GAN	0.55	0.00	3.33	0.00
MAE-GAN with Loss	**92.22**	**83.33**	**96.11**	**94.44**

B. *MAE-GAN Results*

The loss function for MAE-GAN is

$$\begin{aligned} L_{\text{gen}} =& L_{\text{recon}} + \alpha_1 \cdot L_{\text{adv\_image}} + \alpha_2 \cdot L_{\text{adv\_patch}} + \\ & \alpha_3 \cdot L_{\text{percep}} + \alpha_4 \cdot L_{\text{bound}} \end{aligned}$$

We conducted an experiment comparing three model configurations:

1) *Original MAE* as: $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$.
2) *MAE-GAN without perceptual or boundary losses* as: $\alpha_1 = \alpha_2 = 0.2,\ \alpha_3 = \alpha_4 = 0$.
3) *MAE-GAN with perceptual and boundary losses* as: $\alpha_1 = \alpha_2 = 0.5,\ \alpha_3 = 1.5,\ \alpha_4 = 2.0$.

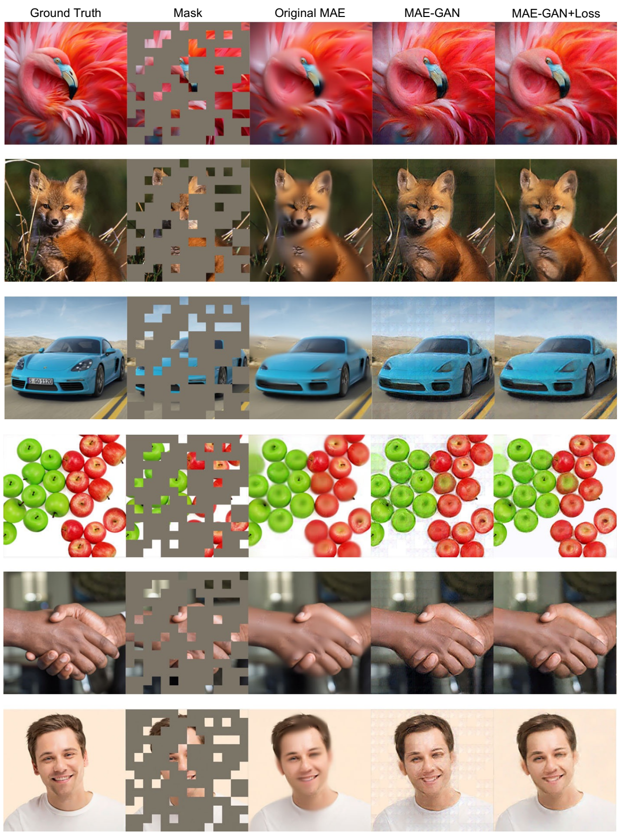

Fig. 4. **MAE-GAN Reconstructed Images.** For each case, from left to right are ground truth, mask, inpainting result by MAE, inpainting result by MAE-GAN without perceptual or boundary losses, and inpainting result by MAE-GAN with perceptual and boundary losses. We show the predicted patches combined with the unmasked patches.

We assessed the quality of images generated by MAE as well as our models from both perspectives: qualitative and quantitative.

Qualitative Assessment. We evaluated the image inpainting performance of all three models by case studies and manual evaluation. Fig. 3 presents cases for those three models. The visual quality of the images varies significantly. The original MAE generates overly-smooth images that lack fine details, as it relies solely on pixel-level reconstruction. The MAE-GAN without perceptual or boundary losses can generate images with improved clarity, but the results exhibit noticeable block-like artifacts and inconsistency at patch boundaries. In contrast, the MAE-GAN with perceptual and boundary losses achieves the best results, producing sharp, detailed reconstructions while not demonstrating significant undesired block artifacts or patch-patch inconsistency.

We further annotated 60 images generated by the original MAE, MAE-GAN, and MAE-GAN with perceptual and boundary losses. For each image case, we evaluated the best image from the following four perspectives:

1) *Overall*: The most visually appealing image is selected among the three options.
2) *Grid Artifact*: Annotators select the image with the least noticeable grid-like artifact.
3) *Blurriness Artifact*: The annotators choose the least blurry image.
4) *Image-Patch Artifact*: The image with the smoothest transition is selected by annotators between model-generated patches and unmasked image patches.

All three authors participated in the manual annotation, and the results are shown in the Tab. II. For all four human preference metrics, MAE-GAN with our specifically designed loss function achieved overwhelming predilection, which confirms the method's ability to generate images with high human preference.

Quantitative Assessment. Our quantitative assessment involves two metrics: LPIPS Score (Perceptual Similarity) and FID Score.

1) *Learned Perceptual Image Patch Similarity (LPIPS)* [13], often referred to as "Perceptual Similarity", is a metric used to quantitatively measure the perceptual similarity between images and is widely used in computer vision tasks. By comparing images in the high-level latent feature space, this metric aligns with human perception and human preference. When LIPIS is measured on two groups of images, a lower score indicates the two groups of images are more

similar or have more similar statistics. A perfect score of 0.0 indicates that the two groups of images are identical.

2) *Fréchet inception distance (FID)* [14] is first used in GANs to measure the realism and diversity of generated images. It combines the assessment of both visual quality and diversity into a single metric. A lower FID score indicates that the images produced by a generative image model are more realistic.

TABLE II
QUANTITATIVE RESULTS FOR THREE IMAGE INPAINTING MODELS. THE HIGHEST ENTRY IS BOLDED. "MAE-GAN WITH LOSS" REFERS TO THE MODEL TRAINED WITH OUR CUSTOM-DESIGNED LOSS FUNCTION. FOR BOTH METRICS, NAMELY THE LPIPS AND FID SCORES, LOWER IS BETTER. THE BEST ENTRIES ARE BOLDED.

Model Architecture	LPIPS Score	FID Score
MAE	0.3235	156.79
MAE-GAN	0.3391	122.10
MAE-GAN with Loss	**0.3156**	**112.60**

We compared 60 inpainted images with their corresponding ground truth and calculated the LPIPS and FID scores between each pair of inpainted and ground truth images. The MAE-GAN model with the novel loss function achieves the best results for both metrics. In the LPIPS score, it surpasses the suboptimal MAE-GAN with conventional loss function by 0.024. It also achieved the best FID score of 112.60, which surpassed the baseline MAE model, and the MAE model with conventional loss function by 44.19 and 9.5, respectively. Overall, the MAE-GAN trained with the novel loss function demonstrates a significant advantage over two competing models on both quantitative metrics, showing the effectiveness of our approach in generating images that are realistic and human-preferable.

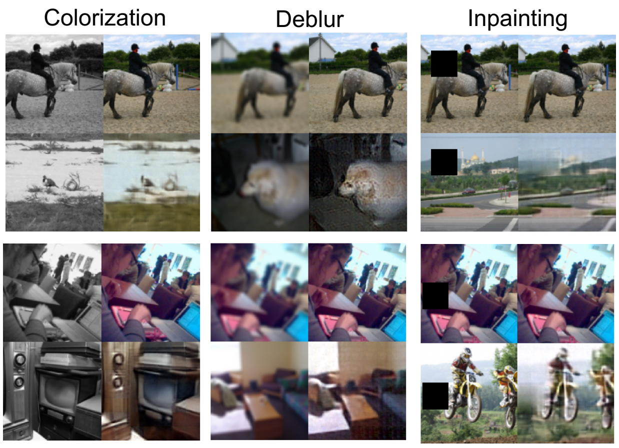

Fig. 5. **Visual Prompting Results by MAE-GAN.** The bottom-right quadrant of each image lies the MAE-GAN prediction. The fourth quadrant is the generated image.

C. *Visual Prompting Results*

We evaluated MAE-GAN's ability to perform visual prompting on three tasks: grayscale image colorization, image deblurring, and image inpainting. Fig. 5 shows some resulting images for the three tasks. Our model demonstrates strong performance in colorization, effectively restoring realistic colors from the provided grayscale images. Similarly, our model proved to be effective in deblurring tasks and managed to restore some degree of sharpness to images degraded by Gaussian blur. However, we observed suboptimal performance in image inpainting, where the masked region covers 10% of the image. Overall, the qualitative results demonstrate that a single MAE-GAN model is capable of handling various visual tasks effectively.

V. DISCUSSION

Image Inpainting. The models capture or miss the same objects in the reconstruction. Interestingly, in Fig. 4, row 4, the model successfully reconstructs the red apple that is being fully masked but hallucinates a green apple that is not shown in the ground truth image. In Fig. 4, row 5, the models fail to reconstruct fully masked fingers, resulting in implausible inpainting where the defects are particularly noticeable.

This behavior highlights a key challenge for MAE-GAN: the ambiguity of fully masked regions. When an object is entirely masked, the models often diverge from the ground truth because they must rely solely on contextual cues or learned priors to guess the content. This can lead to results that are inconsistent with reality, such as hallucinated items that were not present or the omission of crucial details. Furthermore, when fully masked regions are intended to be erased, the models may interpret this differently: either retaining traces of the original item or generating entirely new content to fill the space.

Visual Prompting. The visual prompting technique effectively unlocks the multi-task potential of our image inpainting model. Although our model can hypothetically perform any tasks on which it has been trained with visual prompting, it does not perform equally well on all tasks. This reveals that the effectiveness of visual prompting depends on the complexity of the task. Certain tasks inherently align better with the model's architecture and learning dynamics, while others expose limitations in its generalization or reconstruction capabilities.

VI. FUTURE WORK

In our experiments, we fine-tuned only the MAE decoder and the discriminator. One future direction is to fine-tune the entire MAE-GAN architecture, including the MAE encoder, to achieve higher-quality results. Another future direction is to extend MAE-GAN to handle irregular masks. The current vision transformer backbone cannot process irregular masks directly due to its reliance on masks composed of square patches. One naive solution is to enlarge irregular masks to jagged, patch-aligned masks compatible with MAE-GAN, and paste the reconstruction back onto the mask. However, this solution often leads to inconsistencies between the reconstructed and unmasked regions of the image. Future work may consider adding a post-processing network designed to refine inpainting results and ensure a smooth transition between reconstructed and unmasked regions.

VII. Conclusion

In this work, we enhanced MAE-GAN, a masked Autoencoder integrated into a GAN-like architecture. The generator and discriminator were designed with specialized loss functions to improve image clarity while minimizing unwanted artifacts. The performance of MAE-GAN was evaluated by both quantitative metrics and qualitative analysis, demonstrating its ability to produce clear images with minimal artifacts. Furthermore, we incorporated a visual prompting technique, enabling MAE-GAN to handle multiple visual tasks. These findings highlight MAE-GAN as a promising framework that not only improves MAE's performance in image inpainting but also generalizes across diverse generative tasks.

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